LINEAR ALGEBRA · LESSON 13

Orthogonal Vectors and Subspaces

Orthogonal = perpendicular. In markets it means **"has nothing to do with"**: market-neutral, uncorrelated.

$x^Ty = 0$ | correlation = cos θ | MIT 18.06, Lecture 14

THE TEST

Dot product zero

$$x \perp y \iff x^T y = 0$$

Why: Pythagoras. $\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2x^T y$, so the right-triangle rule holds exactly when $x^T y = 0$.

Zero-cost trade $(1, -1) \perp$ budget row $(1, 1)$: $1 - 1 = 0$.

IN MARKETS

Correlation is a cosine

Demeaned returns as vectors: Nifty $(-20, 0, 20)$, a strategy $(5, -10, 5)$.

$$(-20)(5) + 0 + (20)(5) = 0$$

Orthogonal = uncorrelated: a market-neutral strategy.

Nifty vs midcap $(-26, 2, 24)$: $\cos\theta = 1000/(28.3 \times 35.4) \approx 0.998$.

THE BIG PICTURE

Two perpendicular pairs

$$C(A^T) \perp N(A) \qquad C(A) \perp N(A^T)$$

$Ax = 0$ means x is perpendicular to **every row**, so to the whole row space.

Dimensions add up: $r + (n - r) = n$, $r + (m - r) = m$. Each is the other's **orthogonal complement.**

WITH FUNDS

Nifty, bonds, balanced

Do-nothing trade $(-0.5, -0.5, 1) \perp$ every rule: return $-6 - 3.5 + 9.5 = 0$.

Reachable targets $\perp$ hidden link $(7, -1, 5)$:

Nifty $7 - 12 + 5 = 0$, bonds $7 - 7 + 0 = 0$.

> SO
>
> A target is reachable $\Leftrightarrow$ it is perpendicular to $(7, -1, 5)$.

Invertible ⇔ independent columns

$N(A^TA) = N(A)$, since $x^TA^TAx = \|Ax\|^2$.

Nifty + bonds: $A^TA = \begin{bmatrix} 146 & 85 \\ 85 & 50 \end{bmatrix}$, det 75: invertible.

Nifty + twin: $\begin{bmatrix} 146 & 146 \\ 146 & 146 \end{bmatrix}$, det 0: singular.

FREDHOLM

A certificate of "impossible"

Either $Ax = b$ has a solution, or some y has $A^T y = 0$ and $y^T b = 1$.

Spread loop $y = (1, 1, -1)$, quotes $(100, 150, 240)$: $A^T y = 0$, $y^T b = 10$.

Proof the quotes can't come from real prices: an **arbitrage**.

RECAP

Key takeaways

1. Orthogonal $\Leftrightarrow x^T y = 0$.
2. Correlation = cosine of demeaned returns.
3. Row space $\perp$ null space; column space $\perp$ left null space.
4. $A^T A$ invertible $\Leftrightarrow$ independent columns.
5. Fredholm: an impossibility certificate.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A correlation explorer, an orthogonality lab for any matrix, and worked hedging and options examples:

learn.krisarj.com

Based on MIT OpenCourseWare 18.06 Linear Algebra (Prof. Gilbert Strang), used under CC BY-NC-SA 4.0; these slides are shared under the same licence. Not affiliated with or endorsed by MIT. Educational only, not investment advice.